

# **A Proposal for Multi-modal Emotionally Intelligent System**

Team: Paras Khanal, Vincent Sanchez, Shokhina Jalilova

## **Introduction**

Artificial Intelligence (AI) has progressed remarkably in areas such as logical reasoning, pattern recognition, and automation. As AI becomes increasingly integrated into our daily lives, its ability to recognize and respond to human emotions will be essential for meaningful interactions. Emotional Intelligence (EI) in AI is therefore not just a desirable feature, but a necessary next step.

## **Problem Statement**

Present-day AI systems are capable of exchanging information and simulating rational dialogue, but frequently lack the capacity to recognize or react sympathetically to human emotions. For example, current mental health chatbots can provide scripted responses but fail to detect when a user is genuinely distressed versus merely reporting a bad day. This lessens the depth and efficacy of communication, particularly in delicate situations like stress management, e-learning, telemedicine, or mental health assistance. AI might not be able to maintain long-term, trust-based interactions with humans without EI. This system can support mental health problems.

## **Deep Learning Task/ Motivation & Objectives**

In this project, a machine learning model will be trained to classify the emotions expressed in facial expressions and text data. This project aims to embed emotional intelligence into a deep learning system by training the model with available datasets. In addition to using the datasets, by reviewing existing literature, this work will seek to develop a data mining project that can train a model to perform prediction of emotions classification on unseen data.

Deep learning is uniquely suited for this task because:

- CNNs excel at extracting emotional features from facial expressions
- Transformers/RNNs capture contextual and sentimental nuances in text
- Multi-modal deep learning architectures can fuse these complementary signals for more robust emotion recognition than any single modality.

The objectives include, but are not limited to, the following:

- Create a mapping of the emotional states for limited classification labels.
- Determine the most important techniques in the use case, such as multimodal emotion recognition, sentiment analysis, affective computing, and natural language processing.
- Train a model with human emotions.

## **Data**

The initial data for this project will come from a Kaggle dataset called “Emotions Dataset” for text. And for facial images, the dataset is “Facial Emotional Recognition Image Dataset”. We aim

to integrate other datasets to make the model more robust and useful in various scenarios as the project progresses. The link for the initial dataset is provided below:

<https://www.kaggle.com/datasets/bhavikjikadara/emotions-dataset/data>

- About 20,000 samples with 6 emotion classes (sadness, joy, love, anger, fear, surprise).
- CSV with text phrases and labels.

<https://www.kaggle.com/datasets/sujaykapadnis/emotion-recognition-dataset>

- About 35,000 images with 7 emotion classes (angry, disgust, fear, happy, neutral, sad, surprise).
- 48x48 grayscale images.

Since the datasets are not paired, we will treat this as a multi-modal system that can accept either modality independently and fuse predictions, rather than aligned pairs. For training the fusion model, we will create synthetic pairs by combining samples with matching emotion labels.

## **Proposed Methodology**

As a baseline, we will implement a simple, fully connected shallow network which will provide a reference for evaluating the performance of our deep learning and multi-modal models. Moving ahead, our project will employ a PyTorch based CNN for image classification for facial emotion recognition and a Transformer network for text-based emotion recognition. Multi-modal fusion techniques will then be used to integrate the image and text representations for more comprehensive emotion classification.

## **Evaluation Plan**

We will evaluate our models using metrics such as accuracy, a weighted F1-Score (class imbalance), precision/recall per class (which emotions are confused), and a confusion matrix (visualize misclassification patterns). We will also compare our models. We will have model types like baselines (SVM, random forest, shallow CNN), unimodal (image), unimodal (text), and multimodal where we compare within modality performance, across modality performance, fusion gains, and computational efficiency.

## Expected Outcomes

We anticipate that the CNN and Transformer will properly classify emotions from images and text, respectively. The multi-modal model is expected to integrate both sets of information to result in more accurate emotion recognition. Potential challenges include limited data for certain emotions, risk of overfitting from limited data, and the complexity of implementing multi-modal fusion techniques to combine text and image recognition.

## Schedule

The following table includes a tentative schedule for this project:

| <u>Task Name</u>                                | <u>Start Date</u> | <u>Due Date</u> |
|-------------------------------------------------|-------------------|-----------------|
| 1. Project proposal                             | 2/3/2026          | 2/20/26         |
| 2. Data collection                              | 2/21/26           | 3/6/2026        |
| 3. Data preprocessing & exploration             | 3/7/2026          | 3/20/26         |
| 4. Application of data mining algorithms        | 3/21/2026         | 4/3/2026        |
| 5. Comparison of algorithms through experiments | 4/4/2026          | 4/17/2026       |
| 6. Preparation of report and presentation       | 4/18/26           | 4/30/2026       |
| 7. Project presentation                         | 5/1/2026          | 5/1/2026        |

## References

1. Emotions dataset  
<https://www.kaggle.com/datasets/bhavikjikadara/emotions-dataset/data>
2. Facial Emotions recognition Image dataset  
<https://www.kaggle.com/datasets/sujaykapadnis/emotion-recognition-dataset>
3. PyTorch  
<https://pytorch.org/>